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Attitude Estimation Using Light Curves for Space Objects with Unknown Shapes

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Title

Attitude Estimation Using Light Curves for Space Objects with Unknown Shapes

Titel

Attityduppskattning med hjälp av ljuskurvor för rymdobjekt med okända former

Abstract

As the number of Resident Space Objects (RSOs) keeps increasing, Space Situational Awareness (SSA) becomes a critical matter for satellite safety and traffic management. This paper develops a method for attitude estimation of RSOs with unknown shapes using light curves. A ResNet-1D deep neural network is trained to identify eleven simple geometric classes, achieving a validation accuracy of 54%. This classification is followed by an attitude estimation through a Gaussian Process Unscented Kalman Filter (GPUKF). Contrary to usual attitude estimation methods, the GPUKF uses Gaussian Process Regression (GPR) as a non-parametric observation model in order to adapt to RSO's surface properties uncertainties. The simulations show that for specific geometry classes, the network provides high-fidelity attitude estimation, while more complex shapes remains more challenging for classification and inversion.

Sammanfattning

I takt med att antalet rymdobjekt i omloppsbana (Resident Space Objects, RSO) fortsätter att öka blir rymdlägesbild (Space Situational Awareness, SSA) en avgörande fråga för satellitsäkerhet och trafikledning. Denna artikel utvecklar en metod för attitydskattning av RSO med okända former med hjälp av ljuskurvor. Ett djupt neuralt nätverk av typen ResNet-1D tränas för att identifiera elva enkla geometriska klasser och uppnår en valideringsnoggrannhet på 54 %. Denna klassificering följs av en attitydskattning genom ett Gaussian Process Unscented Kalman Filter (GPUKF). Till skillnad från vanliga metoder för attitydskattning använder GPUKF gaussisk processregression (Gaussian Process Regression, GPR) som en icke-parametrisk observationsmodell för att kunna anpassa sig till osäkerheter i RSO:ers ytegenskaper. Simuleringarna visar att nätverket för vissa geometriklasser ger attitydskattning med hög precision, medan mer komplexa former förblir mer utmanande för klassificering och inversion.

FORWARD

This Master Thesis was carried out at Hanada Laboratory, in Kyushu University. I would like to express my deepest gratitude to Professor Toshiya Hanada and Professor Yasuhiro Yoshimura for entrusting me with this subject and for welcoming me into the laboratory. Their experience and guidance were truly valuable throughout my research.

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Matis

NOMENCLATURE

Notations

Symbol	Description
$\mathbf{q}$	Quaternions
$\bar{q}$	Vector part of the quaternion
q_4	Scalar part of the quaternion
$\mathbf{e}$	Principal rotation axis
Φ	Principal rotation angle (rad)
$\mathbf{p}$	Generalized Rodrigues Parameters
$\boldsymbol{\omega}$	Angular velocity (rad$\cdot$ s^{-1})
$\mathbf{J}$	Moment of inertia tensor (kg$\cdot$ m^2)
$\boldsymbol{\tau}$	External torque (N$\cdot$ m)
$\mathbf{h}$	Angular momentum vector (kg$\cdot$m$^2\cdot$s^{-1}) / Bisector vector (BRDF)
ϕ, θ, ψ	Yaw, pitch, roll Euler angles (rad)
$\mathbf{R}_{b/i}$	Rotation matrix from inertial to body-fixed frame
M	Mass of the Resident Space Object (kg)
ρ	Density (kg$\cdot$ m^{-3})
V	Volume of the RSO (m^3)
ΔV	Volume of a tetrahedral simplex (m^3)
w	Scalar triple product
$\mathbf{R}$	Center of mass position vector (m)
P_{jk}	Products of inertia (kg$\cdot$m^2)
f_r	Bidirectional Reflectance Distribution Function (BRDF)
c_d, c_s	Diffuse and specular components of the BRDF
ρ_d	Diffuse reflectance
F_0	Specular reflectance at normal incidence
F	Fresnel reflectance
γ	Combined specular exponent
n_u, n_v	Specular exponents along u_u and u_v
$\mathbf{s}$	Light source direction unit vector
$\mathbf{v}$	Observer direction unit vector
$\mathbf{n}$	Normal unit vector to the facet

$\mathbf{u}_u$, $\mathbf{u}_v$	Anisotropic direction unit vectors
A	Area of a facet (m^2)
f_i	Projection of incident sunlight on facet normal
f_{obs}	Magnitude term per facet
m_{app}	Apparent magnitude
m_{sun}	Apparent magnitude of the Sun (-26.7)
$m_{\text{app,past}}$	Running mean of past apparent magnitudes
r_{obs}	Distance between facet and observer (km)
ε_G	Gaussian detector noise
ε_S	Scintillation noise
σ	Standard deviation of Gaussian noise
$k(\cdot,\cdot)$	Radial basis function kernel
θ_1, θ_2	Kernel hyperparameters
$\boldsymbol{\mu}^*$	GPR predicted mean
$\boldsymbol{\Sigma}^*$	GPR predicted covariance
$\mathscr{D}$	Training dataset
$\mathbf{x}_k$	State vector at time t_k
$\mathbf{y}_k$	Observed signal at time t_k
$\mathbf{x}_{\text{pred}}$, $\mathbf{x}_{\text{obs}}$	Prediction and observation input vectors
χ	Sigma points
W_i	Weights corresponding to the sigma points
λ	UKF scaling parameter
α, κ, β	UKF tuning parameters
$\mathbf{P}$	Covariance matrix
$\mathbf{K}_k$	Kalman gain
$\mathbf{z}$	GAP feature vector
η	Learning rate

Abbreviations

RSO	Resident Space Object
SSA	Space Situational Awareness
STM	Space Traffic Management
LEO	Low Earth Orbit
GEO	Geosynchronous Equatorial Orbit
LTS	Long Term Sustainability
BRDF	Bidirectional Reflectance Distribution Function
GPR	Gaussian Process Regression
GP	Gaussian Process
UKF	Unscented Kalman Filter
UT	Unscented Transformation
GPUKF	Gaussian Process Unscented Kalman Filter
GRP	Generalized Rodrigues Parameters
RBF	Radial Basis Function
ResNet-1D	Residual Network for 1-Dimensional data
CE	Cross-Entropy
Adam	Adaptive Moment Estimation
ReLU	Rectified Linear Unit
BN	Batch Normalization
GAP	Global Average Pooling
RMSE	Root Mean Square Error

TABLE OF CONTENTS

ABSTRACT 2

SAMMANFATTNING (SWEDISH) 2

FORWARD 3

NOMENCLATURE 4

TABLE OF CONTENTS 8

1 INTRODUCTION 9

2 THEORETICAL BACKGROUND 10

2.1 Light Curves 10

2.1.1 Bidirectional Reflectance Distribution Function (BRDF) 10

2.1.2 Light Curves Model 11

2.2 Attitude Estimation 12

2.2.1 Attitude Kinematics and Dynamics of RSOs 12

2.2.2 Inertia of a Polyhedron 14

2.2.3 Gaussian Process Regression 15

2.2.4 Unscented Kalman Filter 16

2.3 Light Curve Inversion 18

2.3.1 ResNet-1D 18

2.3.2 Cross-Entropy Loss Function 19

2.3.3 Adam Optimizer 20

2.3.4 Regularization 20

3 METHODOLOGY 21

3.1 Building a ResNet-1D for Light Curves Inversion 21

3.1.1 Training Data Generation 21

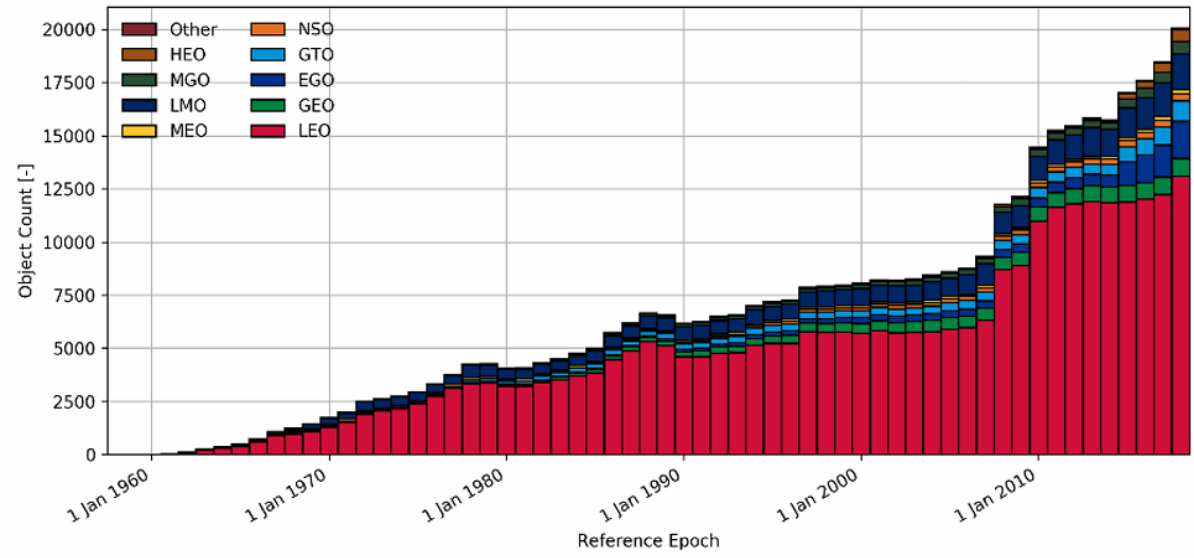
3.1.2 Network Training 24

3.1.3 ResNet-1D Architecture 25

3.2 Attitude Estimation using the GPUKF Algorithm 25
3.2.1 Prediction and Observation Models 25
3.2.2 Data Training 26
3.2.3 GPUKF Algorithm 27
4 RESULTS 28
4.1 Shape Modeling 28
4.2 Attitude Estimation 30
5 DISCUSSION 33
6 CONCLUSION AND FUTURE WORK 34
7 REFERENCES 35
APPENDIX A: UNDER-FIT ρ_D CASE 37
APPENDIX B: OVER-FIT ρ_D CASE 39

1 INTRODUCTION

For the past 20 years, there has been a really important increase in the population of resident space objects (RSOs); especially in Low Earth Orbit (LEO), as shown in Figure 1.



Source: European Space Agency. 2017

Figure 1. Count Evolution by Object Orbit [1]

With this drastically increase, space can no longer be considered "too big" for any satellite mission to not take into account the debris. Thus, Space Traffic Management (STM) has become a serious issue, ensuring the safety, sustainability, and stability of all space activity. STM lies on Space Situational Awareness (SSA), which is defined as the knowledge, characterization, and practice of tracking of space objects and their operational environment [2].

All SSA data can be used for any space activity to prevent collisions with RSOs and to decide possible avoiding maneuvers in the case of debris coming too close to the decided path, thus becoming a foundation for STM [3].

This paper will focus on attitude estimation of unknown shaped RSOs using light curves. For many RSOs, direct imaging remains impossible, making photometric data a necessity for parametric characterization, and essentially attitude as studied in the paper. Attitude estimation is fundamental to model the accurate propagation of an RSO's orbit, since gravitational and non gravitational forces, such as gravity gradients, solar radiation pressure and atmospheric drag, depend on the object's attitude.

While traditional methods often rely on a-priori knowledge of the objects' geometry or parameters, unknown objects are studied with really complex algorithms and deep networks ([4], [5]). This study uses a Residual Network for 1-Dimensional (ResNet-1D) data to classify different shapes, and a Gaussian Process (GP) to handle the surface property uncertainties. This GP is associated with a typical Unscented Kalman Filter (UKF) to finally estimate the attitude of RSOs.

2 THEORETICAL BACKGROUND

2.1 Light Curves

2.1.1 Bidirectional Reflectance Distribution Function (BRDF)

Let us consider a geometry as shown in Figure 2. The geometry is described as a right handed frame using the unit vectors $\mathbf{u}_u$, $\mathbf{u}_v$ and $\mathbf{n}$: $\mathbf{u}_u$ and $\mathbf{u}_v$ being the anisotropic directions and $\mathbf{n}$ normal to the facet. We also need to consider $\mathbf{s}$ and $\mathbf{v}$, respectively, unit vectors of the light source direction and the observer direction. Both of these vectors can be expressed locally using the frame azimuth (ϕ_i, θ_i) and elevation (ϕ_r, θ_r) angles.

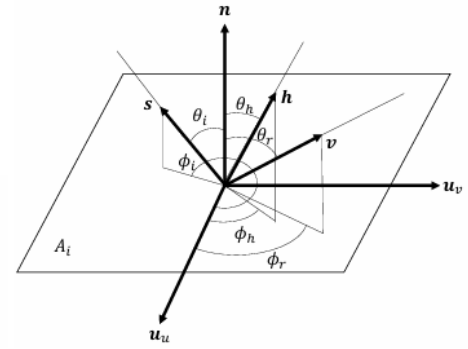


Figure 2. Geometry of the light reflection on a facet [6]

The bisector vector $\mathbf{h}$ between the light source direction $\mathbf{s}$ and observer direction $\mathbf{v}$ is defined as:

$$\mathbf{h} = \frac{\mathbf{s}+\mathbf{v}}{||\mathbf{s}+\mathbf{v}||} = \begin{bmatrix} \sin\theta_h \cos\phi_h \\ \sin\theta_h \sin\phi_h \\ \cos\alpha \end{bmatrix} \tag{1}$$

A BRDF f_r is used to describe the reflection property of a complex shaped opaque surface and can then be defined as [7]:

$$f_r(\mathbf{s},\mathbf{v}) = \frac{\mathrm{d}L_r(\mathbf{v})}{\mathrm{d}E_i(\mathbf{s})} = \frac{\mathrm{d}L_r(\mathbf{v})}{\mathrm{d}L_i(\mathbf{s})\cos(\theta_i)\mathrm{d}s} \tag{2}$$

where L is the radiance along the incident or reflected direction, E is the irradiance.

This form will not be used in this paper but the BRDF will be represented as a sum of diffuse and specular components:

$$f_r = c_d + c_s \tag{3}$$

Using the Ashikhmin–Shirley model [8], the diffusive and specular components are written as followed:

$$c_d = \frac{28\rho_d}{23\pi}(1-F_0)\left[1-\left(1-\frac{\mathbf{n}^T\mathbf{s}}{2}\right)^5\right]\left[1-\left(1-\frac{\mathbf{n}^T\mathbf{v}}{2}\right)^5\right] \tag{4}$$

$$c_s = \frac{\sqrt{(n_u+1)(n_v+1)}}{8\pi}\frac{F}{\mathbf{v}^T\mathbf{h}\max\{\mathbf{n}^T\mathbf{s},\mathbf{n}^T\mathbf{v}\}}(\cos\alpha)^\gamma \tag{5}$$

The exponent γ and the Fresnel reflectance F are expressed as followed:

$$\gamma = n_u\cos^2\phi_h + n_v\sin^2\phi_h = \frac{n_u(\mathbf{h}^T\mathbf{u}_u)^2 + n_v(\mathbf{h}^T\mathbf{u}_v)^2}{1-(\mathbf{h}^T\mathbf{n})^2} \tag{6}$$

$$F = F_0 + (1-F_0)(1-\mathbf{v}^T\mathbf{h})^5 \tag{7}$$

The Ashikhmin–Shirley model then has four different parameters, being ρ_d the color that specifies the diffuse reflectance of the "substrate" under the specular coating, F_0 the color that specifies the specular reflectance at normal incidence and, n_u and n_v respectfully controlling the shape of the specular lobe along $\mathbf{u}_u$ and $\mathbf{u}_v$.

2.1.2 Light Curves Model

A light curve represents the brightness of an object over time. Considering a facet with the geometry in Figure 2, we can define a magnitude term relative to each facet as:

$$f_{\text{obs}} = f_r f_i A(\mathbf{n}^T\mathbf{v}) \tag{8}$$

where A is the area of the facet, f_r is the BRDF and f_i is the projection of the incident sunlight on the normal $\mathbf{n}$.

$$f_i = \begin{cases} 0 & (\mathbf{n}^T\mathbf{s} \le 0) \\ \mathbf{n}^T\mathbf{s} & (\mathbf{n}^T\mathbf{s} > 0) \end{cases} \tag{9}$$

Considering an RSO as a junction of many small facets, the apparent magnitude of the brightness of an RSO observed on the ground is then written as:

$$m_{\text{app}} = m_{\text{sun}} - 2.5\log_{10}\left|\sum_{\text{facets}}\frac{f_{\text{obs}}}{r_{\text{obs}}^2}\right| \tag{10}$$

where $m_{\text{sun}} = -26.7$ is the apparent magnitude of the brightness of the Sun, f_{obs} the magnitude term in Equation (8), and r_{obs} is the distance separating the facet and the observer. Since the distance between the RSO and the observer will be much greater than the dimension of the RSO, r_{obs} can be considered equal for each facet.

2.2 Attitude Estimation

2.2.1 Attitude Kinematics and Dynamics of RSOs

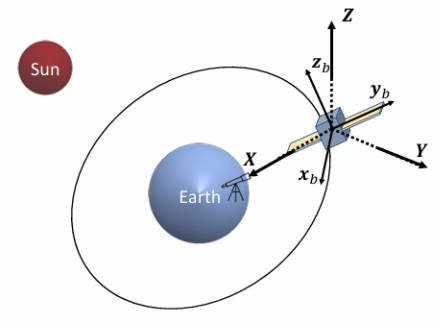


Figure 3. XYZ coordinate system and body-fixed frame [6]

For this paper, we will consider a coordinate system and frame as shown in Figure 3. The XYZ pseudo-inertial coordinate system, with a fixed positioned Sun, Earth and observer, is well adapted to describe the position of RSOs orbiting around. In this measure, we choose to consider the center of mass of the RSO we study as the origin of the coordinate system. The X axis is aimed towards the observer, the Z axis along with the North pole of the Earth, and the Y axis is defined so that it forms a right-handed frame [6].

We also need to consider a (x_b, y_b, z_b) body-fixed frame originated at the center of mass of the RSO, with each axis directed along the principal axes of inertia . This body-fixed frame is used to express the RSO attitude [6].

For attitude representations, quaternions [9] will be used, and defined as follows:

$$\mathbf{q} = [q_1, q_2, q_3, q_4]^T = [\bar{q}^T, q_4]^T = \begin{bmatrix} \mathbf{e} \sin \frac{\Phi}{2} \\ \cos \frac{\Phi}{2} \end{bmatrix} \tag{11}$$

where $\bar{q} = [q_1, q_2, q_3]$ is called the vector part and q_4 the scalar part of the quaternions. Quaternions are defined using the principal axis $\mathbf{e}$ and a principal angle Φ as represented in Figure 4.

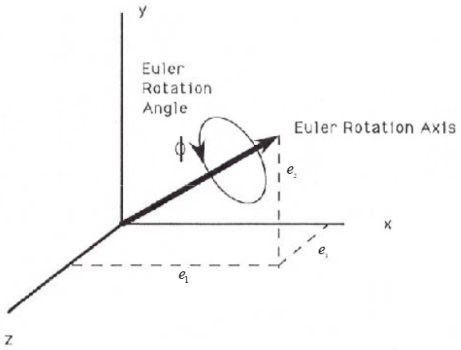


Figure 4. Principal rotation vector [9]

Quaternions major advantage is that they do not have any singularity. However, they must follow a norm constraint at all times:

$$\mathbf{q}^T\mathbf{q} = q_1^2 + q_2^2 + q_3^2 + q_4^2 = 1 \tag{12}$$

This constraint results in an inability to use regular arithmetic operations when calculating the attitude with quaternions. Thus, a quaternion product is defined to represent a change of attitude:

$$\mathbf{q} \otimes \mathbf{q'} = \begin{bmatrix} q_4\bar{q}' + q_4'\bar{q} - \bar{q} \times \bar{q}' \\ q_4 q_4' - \bar{q}^T\bar{q}' \end{bmatrix} \tag{13}$$

The kinematics of the quaternion q is written as:

$$\frac{d}{dt}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} = \frac{1}{2}\tilde{\omega} \otimes \mathbf{q} \Leftrightarrow \begin{bmatrix} \dot{q_1} \\ \dot{q_2} \\ \dot{q_3} \\ \dot{q_4} \end{bmatrix} = \frac{1}{2}\begin{bmatrix} 0 & \omega_3 & -\omega_2 & \omega_1 \\ -\omega_3 & 0 & \omega_1 & \omega_2 \\ \omega_2 & -\omega_1 & 0 & \omega_3 \\ -\omega_1 & -\omega_2 & -\omega_3 & 0 \end{bmatrix}\begin{bmatrix} q_1 \\ q_2 \\ q_3 \\ q_4 \end{bmatrix} \tag{14}$$

where $\boldsymbol{\omega}$ is the angular velocity and $\tilde{\boldsymbol{\omega}} = [\boldsymbol{\omega}^T, 0]^T$.

Since they have no constraints, unlike quaternions, generalized Rodrigues parameters (GRP) are also used in this paper for a weighted average of the sigma points in the UKF algorithm. They are defined as:

$$\mathbf{p} = f\frac{\bar{q}}{a + q_4} \tag{15}$$

One other issue with using a quaternion representation is the difficulty in trying to visualize the attitude. To solve this problem, an Euler angle representation will be used. Those angles describe the orientation between two frames using generally three sequential rotations around specific axes.

For this paper, we will consider an asymmetric sequence (3-2-1) yaw (ϕ), pitch (θ), roll (ψ). This means the following successive rotations $\mathbf{R}_z(\phi)$, $\mathbf{R}_y(\theta)$, and $\mathbf{R}_x(\psi)$. This sequence can be represented as one rotation matrix:

$$\begin{aligned} \mathbf{R}_{b/i} &= \mathbf{R}_x(\psi)\mathbf{R}_y(\theta)\mathbf{R}_z(\phi) \\ &= \begin{bmatrix} \cos\theta\cos\psi & \cos\theta\sin\psi & -\sin\theta \\ \sin\theta\sin\phi\cos\psi - \cos\phi\sin\psi & \sin\theta\sin\phi\sin\psi + \cos\phi\cos\psi & \cos\theta\sin\phi \\ \sin\theta\cos\phi\cos\psi + \sin\phi\sin\psi & \sin\theta\cos\phi\sin\psi - \sin\phi\cos\psi & \cos\theta\cos\phi \end{bmatrix} \end{aligned} \tag{16}$$

Quaternions can be expressed using Euler angles as:

$$\mathbf{q} = \begin{bmatrix} \cos\frac{\phi}{2}\cos\frac{\theta}{2}\sin\frac{\psi}{2} + \sin\frac{\phi}{2}\sin\frac{\theta}{2}\cos\frac{\psi}{2} \\ \cos\frac{\phi}{2}\sin\frac{\theta}{2}\cos\frac{\psi}{2} - \sin\frac{\phi}{2}\cos\frac{\theta}{2}\sin\frac{\psi}{2} \\ \cos\frac{\phi}{2}\sin\frac{\theta}{2}\sin\frac{\psi}{2} + \sin\frac{\phi}{2}\cos\frac{\theta}{2}\cos\frac{\psi}{2} \\ \cos\frac{\phi}{2}\cos\frac{\theta}{2}\cos\frac{\psi}{2} - \sin\frac{\phi}{2}\sin\frac{\theta}{2}\sin\frac{\psi}{2} \end{bmatrix} \tag{17}$$

RSOs are assumed to be rigid bodies, which attitude motion can be represented using Euler equation of motion. As a result of the conservation of the angular momentum, this equation can then be written as:

$$\frac{\mathrm{d}\mathbf{h}}{\mathrm{d}t} = \frac{\mathrm{d}^*\mathbf{h}}{\mathrm{d}t} + \boldsymbol{\omega} \times \mathbf{h} = \mathbf{J}\frac{\mathrm{d}^*\boldsymbol{\omega}}{\mathrm{d}t} + \boldsymbol{\omega} \times \mathbf{J}\boldsymbol{\omega} = \boldsymbol{\tau} \tag{18}$$

where $\mathbf{h}$ is the vector of angular momentum, J is the moment of inertia, $\boldsymbol{\tau}$ is the external torque, and $\frac{\mathrm{d}}{\mathrm{d}t}$ and $\frac{\mathrm{d}^*}{\mathrm{d}t}$ are respectively the time differential with respect to the inertial frame and the body-fixed frame.

2.2.2 Inertia of a Polyhedron

In this paper, a triangle subdivision is applied to every modeled shape, making the calculation of the inertia [10] easier. This representation is illustrated as an example in Figure 5.

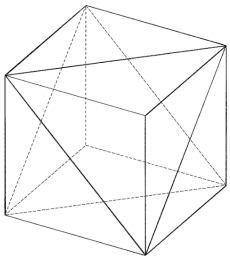


Figure 5. Subdivision of a cube into triangle facets [10]

Let a facet have vertices D, E, and F ordered counter-clockwise as seen from outside. Let **D**, **E**, and **F** be their associated vector, with the origin O as represented in Figure 6.

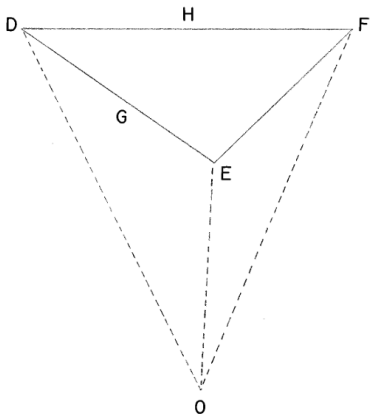


Figure 6. Geometry representation of a facet (simplex) [10]

Using this representation, the volume of the tetrahedron is defined by:

$$\Delta V = \frac{w}{6} \tag{19}$$

$$w = \mathbf{D} \cdot (\mathbf{E} \times \mathbf{F}) \tag{20}$$

where w is the scalar triple product. The volume of the RSO is then the sum of the volume of all faces:

$$V = \sum \Delta V = \frac{1}{6} \sum w \tag{21}$$

Another crucial information when calculating the inertia of a body, is finding its center of mass R. For a single simplex, the centroid is defined as:

$$\Delta \mathbf{R} = \frac{\mathbf{D} + \mathbf{E} + \mathbf{F}}{4} \tag{22}$$

To extract the position of the center of mass, each simplex centroid is weighted by its corresponding volume contribution:

$$\mathbf{R} = \sum \frac{w \cdot (\mathbf{D} + \mathbf{E} + \mathbf{F})}{4 \sum w} \tag{23}$$

Combining equation (23) with (21), it gives:

$$\mathbf{R} = \frac{1}{24V} \sum w \cdot (\mathbf{D} + \mathbf{E} + \mathbf{F}) \tag{24}$$

Considering an RSO of mass M and density ρ, its rotational inertia can be expressed as:

$$\mathbf{J} = \begin{bmatrix} P_{yy} + P_{zz} & -P_{xy} & -P_{xz} \\ -P_{xy} & P_{xx} + P_{zz} & -P_{yz} \\ -P_{xz} & -P_{yz} & P_{xx} + P_{yy} \end{bmatrix} \tag{25}$$

where $P_{jk} = \sum \Delta P_{jk}$ are the products of inertia expressed as:

$$\Delta P_{jk} = \frac{\rho w}{120} (2D_j D_k + 2E_j E_k + 2F_j F_k + D_j E_k + D_k E_j + D_j F_k + D_k F_j + E_j F_k + E_k F_j) \tag{26}$$

However, it is more common to express the inertia tensor about the RSO's center of mass $\mathbf{R} = (X, Y, Z)$. Thus, the parallel axis theorem gives:

$$\mathbf{J}' = \mathbf{J} - M \begin{bmatrix} Y^2 + Z^2 & -XY & -XZ \\ -XY & X^2 + Z^2 & -YZ \\ -XZ & -YZ & X^2 + Y^2 \end{bmatrix} \tag{27}$$

with $\mathbf{J}$ the inertia tensor in Equation (25).

2.2.3 Gaussian Process Regression

The Gaussian process regression (GPR) method ([6], [11], [12]) is used to model non-linear systems with no knowledge of parameters. The goal of the GPR is to learn about an unknown function defined by $\mathbf{y}_{gp} = GP(\mathbf{x}_{gp})$, $\mathbf{x}_{gp}$ being the input feature vector and $\mathbf{y}_{gp}$ the observed

target corresponding. The GPR will predict the mean expected output as well as a confidence interval.

Having $\mathbf{X}_{gp} = [\mathbf{x}_{gp,1}, \mathbf{x}_{gp,2}, \ldots, \mathbf{x}_{gp,n}]$ and $\mathbf{Y}_{gp} = [\mathbf{y}_{gp,1}, \mathbf{y}_{gp,2}, \ldots, \mathbf{y}_{gp,n}]$, a training data set is defined as:

$$\mathscr{D} = \langle \mathbf{X}_{gp}, \mathbf{Y}_{gp} \rangle = \{(\mathbf{x}_{gp,1}, \mathbf{y}_{gp,1}), \ldots, (\mathbf{x}_{gp,n}, \mathbf{y}_{gp,n})\} \tag{28}$$

The output vector $\mathbf{Y}_{gp}$ is set to be Gaussian:

$$\mathbf{Y}_{gp} \sim \mathscr{N}(0, \mathbf{K}) \tag{29}$$

where $\mathbf{K} \in \mathbb{M}_{n \times n}$ is the covariance matrix defined with a kernel function as $K_{i,j} = k(x_{gp,i}, x_{gp,j})$ for $i, j = 1, 2, \ldots, n$. The kernel function defines the similarity between vectors and is set to be a radial basis function (RBF) kernel in this paper:

$$k(x_{gp,i}, x_{gp,j}) = \theta_1 \exp\left(-\frac{\left\|x_{gp,i} - x_{gp,j}\right\|^2}{\theta_2}\right) \quad i, j = 1, 2, \ldots, n \tag{30}$$

with $\theta_1 = 1$ and $\theta_2 = 0.25$, two hyperparameters of the kernel function.

Considering a new input vector $\mathbf{x}^*_{gp}$, posterior statistics can be defined as:

$$\boldsymbol{\mu}^* = \mathbf{K}_*^T \mathbf{K}^{-1} \mathbf{Y}_{gp} \tag{31}$$
$$\boldsymbol{\Sigma}^* = \mathbf{K}_{**} - \mathbf{K}_*^T \mathbf{K}^{-1} \mathbf{K}_* \tag{32}$$

$$\mathbf{K}_* = k(\mathbf{X}_{gp}, \mathbf{x}^*_{gp}) \tag{33}$$
$$\mathbf{K}_{**} = k(\mathbf{x}^*_{gp}, \mathbf{x}^*_{gp}) \tag{34}$$

where $\boldsymbol{\mu}^*$ and $\boldsymbol{\Sigma}^*$ are the predicted statistics of the output vector $\mathbf{y}^*_{gp} = GP(\mathbf{x}^*_{gp})$:

$$p(\mathbf{y}^*_{gp} | \mathbf{x}^*_{gp}, \mathscr{D}) = \mathscr{N}(\boldsymbol{\mu}^*, \boldsymbol{\Sigma}^*) \tag{35}$$

2.2.4 Unscented Kalman Filter

The unscented Kalman filter (UKF) allows us to do a state estimation on a non-linear dynamic system ([6], [13], [14]). Let us then consider the following discrete time non-linear system:

$$\mathbf{x}_{k+1} = F(t_k, \mathbf{x}_k, \mathbf{v}_k, \mathbf{u}_k) \in \mathbb{R}^L \tag{36}$$
$$\mathbf{y}_k = H(t_k, \mathbf{x}_k, \mathbf{n}_k, \mathbf{u}_k) \in \mathbb{R}^m \tag{37}$$

where $\mathbf{x}_k$ is the unobserved state vector of the system, $\mathbf{y}_k$ is the only observed signal at time t_k, $\mathbf{u}_k$ is the input vector, $\mathbf{v}_k$ and $\mathbf{n}_k$ are respectively process noise driving the dynamic system and

observation noise, and F and H are the system dynamic model supposed to be known. Equation (36) is called the prediction model and Equation (37) is called the observation model.

To go through the UKF, the unscented transformation (UT) first needs to be explained. This method is used for calculating the statistics of random variables that undergo a non-linear transformation, as it is the case for the system above.

Let $\hat{\mathbf{x}}_k$ and $\mathbf{P}_k$ be the mean and the covariance matrix of the estimated state $\mathbf{x}_k$ at time t_k respectively. Sigma points are expressed as:

$$\boldsymbol{\chi}_{k-1,0} = \hat{\mathbf{x}}_{k-1} \tag{38}$$

$$\boldsymbol{\chi}_{k-1,i} = \boldsymbol{\chi}_{k-1,0} + \sqrt{(L+\lambda)\mathbf{P}_{k-1,i}} \quad (i = 1,2,\ldots,L) \tag{39}$$

$$\boldsymbol{\chi}_{k-1,i} = \boldsymbol{\chi}_{k-1,0} - \sqrt{(L+\lambda)\mathbf{P}_{k-1,i}} \quad (i = L+1,\ldots,2L) \tag{40}$$

$$\lambda = \alpha^2(L+\kappa) - L \tag{41}$$

λ is a scaling parameter with L the dimension of the estimated state vector, $\alpha = 10^{-4}$ showing the spread of the sigma points around $\hat{x}_{k-1}$, and $\kappa = 3 - L$ a scaling parameter.

Corresponding weights W_i are also defined as:

$$W_{0,\text{mean}} = \frac{\lambda}{L+\lambda} \tag{42}$$

$$W_{0,\text{cov}} = \frac{\lambda}{L+\lambda} + (1 - \alpha^2 + \beta) \tag{43}$$

$$W_{\text{i,mean}} = W_{\text{i,cov}} = \frac{1}{2(L+\lambda)} \quad (i = 1,2,\ldots,2L) \tag{44}$$

where β is a parameter used to incorporate knowledge of the distribution of $\hat{\mathbf{x}}_k$ and can be set equal to 2 for a Gaussian distribution.

Using the prediction model (36) with the sigma points, predicted sigma points can be expressed as:

$$\boldsymbol{\chi}_{k|k-1} = F(t_{k-1}, \boldsymbol{\chi}_{k-1}, \mathbf{v}_{k-1}, \mathbf{u}_{k-1}) \tag{45}$$

The a-priori estimated state and covariance matrix are then calculated as a weighted average of the predicted sigma points:

$$\hat{\mathbf{x}}_k^- = \sum_{i=0}^{2L} W_{\text{i,mean}} \boldsymbol{\chi}_{k|k-1,i} \tag{46}$$

$$\mathbf{P}_k^- = \sum_{i=0}^{2L} W_{\text{i,cov}} (\boldsymbol{\chi}_{k|k-1,i} - \hat{\mathbf{x}}_k^-)(\boldsymbol{\chi}_{k|k-1,i} - \hat{\mathbf{x}}_k^-)^T \tag{47}$$

Combining Equations (37) and (45) with a weighted average similarly to above, the a-priori predicted observation is:

$$\mathscr{Y}_k^- = H(t_k, \boldsymbol{\chi}_{k|k-1}, \mathbf{n}_k, \mathbf{u}_k) \tag{48}$$

$$\hat{\mathbf{y}}_k^- = \sum_{i=0}^{2L} W_{\text{i,mean}} \mathscr{Y}_{k,i}^- \tag{49}$$

To compute the posterior estimated value $\hat{\mathbf{x}}_k^+$ and covariance matrix $\mathbf{P}_k^+$, the a-priori expression needs to be updated with the Kalman gain defined as:

$$\mathbf{K}_k = \mathbf{P}_{x_k y_k} \mathbf{P}_{y_k y_k}^{-1} \tag{50}$$

with $\mathbf{P}_{x_k y_k}$ and $\mathbf{P}_{y_k y_k}$ covariance matrices:

$$\mathbf{P}_{x_k y_k} = \sum_{i=0}^{2L} W_{\text{i,cov}} (\boldsymbol{\chi}_{k|k-1,i} - \hat{\mathbf{x}}_k^-)(\mathscr{Y}_{k,i}^- - \hat{\mathbf{y}}_k^-)^T \tag{51}$$

$$\mathbf{P}_{y_k y_k} = \sum_{i=0}^{2L} W_{\text{i,cov}} (\mathscr{Y}_{k,i}^- - \hat{\mathbf{y}}_k^-)(\mathscr{Y}_{k,i}^- - \hat{\mathbf{y}}_k^-)^T \tag{52}$$

The UKF update equations are finally computed as:

$$\hat{\mathbf{x}}_k^+ = \hat{\mathbf{x}}_k^- + \mathbf{K}_k(\mathbf{y}_k - \hat{\mathbf{y}}_k^-) \tag{53}$$

$$\mathbf{P}_k^+ = \mathbf{P}_k^- - \mathbf{K}_k \mathbf{P}_{y_k y_k} \mathbf{K}_k^T \tag{54}$$

2.3 Light Curve Inversion

Light curves are characterized by measured brightness, which is dependent on the incoming and reflected lights. Thus, the data obtained will differ on the basis of the shape of RSO, its attitude, and other parameters such as its rotation axis or spin. Light curves can then be considered as functions of RSO's parameters and can be inverted to identify those specific parameters.

The inversion problem consists in identifying the RSO parameters through their light curve data by iteratively changing these parameters in a model until both simulated and measured light curves match [15]. This can be mathematically described as:

$$Light_curve = \mathbf{FM}(\theta) + \varepsilon \tag{55}$$

where **FM** is the forward model, ε is the noise process, and θ is the parameter we are trying to identify. This paper will focus on finding the shape of RSOs by training a deep neural network.

2.3.1 ResNet-1D

The residual network for 1D data (ResNet-1D) is a specialized neural network used for reading signals, such as light curves, instead of images.

To build such a deep network, many different filters [15] are used in each step. A convolution filter is defined by:

$$(\mathbf{X} * \mathbf{W})(i) = \sum_{n=0}^{N} X_n * W_{i-n} \tag{56}$$

where $\mathbf{W}$ is the convolution kernel set as random weights and $\mathbf{X}$ is the input data, the light curve in this paper. The kernel goes across the whole curve, identifying any specific temporal signatures.

A rectified linear unit (ReLu) is considered to introduce the nonlinearities of the model:

$$f(x) = \max(0, x) \tag{57}$$

A fundamental aspect of a ResNet-1D is the residual block [16]. In standard deep networks, the gradient signal, which is the information, goes through every convolutional layer during backpropagation. This causes the gradient to become exponentially small before reaching the early layers, preventing them from learning. A residual block creates a "shortcut" letting the original information skip certain layers. Thus, it prevents the disappearance of the gradient.

To improve training, other operations are considered. A batch normalization (BN) consists in normalizing the output of a layer to have a mean of 0 and a variance of 1. This method stabilizes the training process and allows for higher training rates. An other operation is max pooling (MaxPool), used to reduce the spatial dimension of data, while retaining the most important features. To do so, a kernel slides across specific regions of the data and only keeps the maximum value. It allows a reduction of the number of parameters and computational load in the following layers, while focusing on the most prominent signals.

Unlike MaxPool, which focuses on the strongest signal in small windows, global average pooling (GAP) is a layer that looks at a feature map, which is a specific pattern across the data, and calculates its average strength:

$$\mathbf{z} = \frac{1}{L} \sum_{t=1}^{L} h(t) \tag{58}$$

where $\mathbf{z}$ is a feature vector, L is the temporal length entering the GAP layer, and $h(t)$ is the feature map at time t.

The detailed architecture will be described in Section 3.1.3.

2.3.2 Cross-Entropy Loss Function

The network is trained by minimizing a cross-entropy (CE) loss function [15]. Having N samples and C classes, being the different shapes, the loss function is defined as:

$$L_{\mathrm{CE}}(\boldsymbol{\theta}) = -\frac{1}{N} \sum_{i=1}^{N} \sum_{j=1}^{C} y_{ij} \log(\hat{y}_{ij}) \tag{59}$$

where $y_{ij} \in \{0, 1\}$ is the one-hot ground-truth label equal to 1 for the true class of sample i, and $\hat{y}_{ij}$ is the predicted probability of class j. This probability is defined by the softmax function:

$$P(class = j|\mathbf{x}) = \frac{\exp(z_j)}{\sum_{k=1}^{C} \exp(z_k)} \tag{60}$$

with **x** the input light curves and **z** the GAP feature vector from equation (58).

2.3.3 Adam Optimizer

The network parameters $\boldsymbol{\theta}$ are updated using the adaptive moment estimation (Adam) optimizer [17]. First, the gradient g_t of the CE function (59) is calculated:

$$\mathbf{g}_t = \nabla_{\boldsymbol{\theta}} L_{CE}(\boldsymbol{\theta}_t) \tag{61}$$

Adam maintains a moving average of the first moment $\mathbf{m}_t$ and the second moment $\mathbf{v}_t$:

$$\mathbf{m}_t = \beta_1 \mathbf{m}_{t-1} + (1-\beta_1)\mathbf{g}_t \tag{62}$$

$$\mathbf{v}_t = \beta_2 \mathbf{v}_{t-1} + (1-\beta_2)\mathbf{g}_t^2 \tag{63}$$

with β_1 and β_2 decay rates generally set respectively to 0.9 and 0.999.

Both moments are then bias corrected to account for their initialization at zero:

$$\hat{\mathbf{m}}_t = \frac{\mathbf{m}_t}{1-\beta_1^t} \tag{64}$$

$$\hat{\mathbf{v}}_t = \frac{\mathbf{v}_t}{1-\beta_2^t} \tag{65}$$

Finally, the parameters are updated using the corrected moments:

$$\boldsymbol{\theta}_t = \boldsymbol{\theta}_{t-1} - \frac{\eta}{\sqrt{\hat{\mathbf{v}}_t} + \varepsilon} \hat{\mathbf{m}}_t \tag{66}$$

with η the learning rate and ε a small number, generally 10^{-8}, to prevent a division by zero.

2.3.4 Regularization

Three regularization strategies are used to prevent overfitting.

The L2 weight decay consists in adding to the loss function a squared penalty to discourage large weights:

$$L_{\text{total}}(\boldsymbol{\theta}) = L_{\text{CE}}(\boldsymbol{\theta}) + 2 \cdot 10^{-4} ||\boldsymbol{\theta}||^2 \tag{67}$$

The second strategy is the gradient clipping to prevent the gradient from being too big and causing leaps in the network. If its magnitude exceeds a threshold $\tau = 1$, then it is scaled down:

$$\nabla_{\boldsymbol{\theta}} = \nabla_{\boldsymbol{\theta}} \min(1, \frac{\tau}{||\nabla_{\boldsymbol{\theta}}||}) \tag{68}$$

Finally, to improve robustness, a dropout is applied to the GAP feature vector **z** (58). It sets each element of this vector to zero with a probability p, and the non-null elements are scaled by $\frac{1}{1-p}$ to preserve the expected magnitude. In this paper, $p = 0.5$.

3.1 Building a ResNet-1D for Light Curves Inversion

3.1.1 Training Data Generation

The dataset consists of 15000 different light curves. Each input has randomized parameters, allowing the network to learn purely based on shape features and not based on any geometry-specific condition.

First, the physic parameters used to describe the BRDF and the attitude are randomized as shown in Table 1.

Table 1. Physical Model and Scene Parameter Ranges

Category	Parameter	Units	Value / Range
BRDF	Specular Exponents ($n_u = n_v$)	[-]	$U(5,250)$
	Specular Reflectance (F_0)	[-]	$U(0.02,0.14)$
Dynamics	Angular Speed Magnitude ($\|\boldsymbol{\omega}\|$)	[rad/s]	$U(0.001,0.005)$
	Initial Attitude ($\boldsymbol{q}$)	[-]	Random Unit Quaternion
Noise	Gaussian Noise (σ_G)	[-]	0.03
	Scintillation Factor	[-]	$0.015 \cdot \|\mathcal{N}(0,1)\|$

Two noise processes are added to the simulated apparent magnitude (10):

$$m_{\text{obs}}(t) = m_{\text{app}}(t) + \varepsilon_G(t) + \varepsilon_S(t) \tag{69}$$

$$\varepsilon_G(t) \sim \mathcal{N}(0, \sigma_G^2) \tag{70}$$

$$\varepsilon_S(t) = 0.015|m_{\text{app}}(t) - \mu_m| \cdot |\varepsilon'(t)| \quad \varepsilon'(t) \sim \mathcal{N}(0,1) \tag{71}$$

where μ_m is the mean magnitude over the observation window, ε_G is a Gaussian detector noise modeling sensor read noise, and ε_S is a scintillation noise modeling atmospheric turbulence.

Each training sample also uses a specific illumination geometry, forcing the network to learn shape signatures independently of the viewing angle. This geometry is detailed in Table 2.

Table 2. Illumination and Observation Geometry Parameter Ranges

Category	Parameter	Units	Value / Range
Illumination	Sun Azimuth (az_{sun})	[rad]	$U(0,2\pi)$
	Sun Elevation (el_{sun})	[rad]	$U(\frac{\pi}{6},\frac{\pi}{2})$
Sun–Object–Observer Angle	Phase Angle (ψ)	[rad]	$U(\frac{\pi}{6},\frac{2\pi}{3})$
Observation	Observer Azimuth (az_{obs})	[rad]	$az_{\text{sun}} + \psi$
	Observer Elevation (el_{obs})	[rad]	$U(\frac{\pi}{6},\frac{\pi}{2})$

In this paper, eleven specific shape classes are considered as shown in Figure 7. These classes remain of simple geometry due to calculation power limitation.

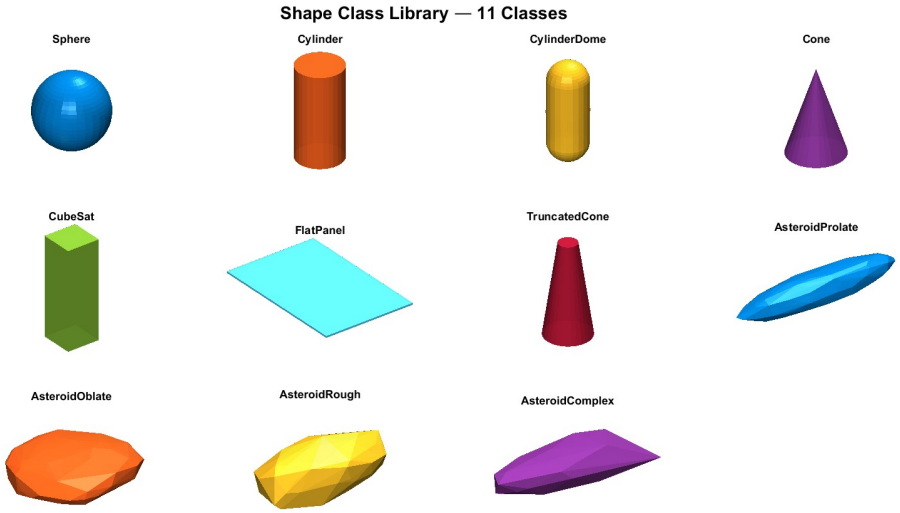


Figure 7. Shape Classes for Training

Shape variation is applied for each light curve, using a randomized version of a class, to ensure that the ResNet-1D learns the archetype and not a specific mesh. This randomization is described in Tables 3 and 4.

Asteroid like shapes have a more complex geometry and can be separated into four different classes depending on their parameter ranges detailed in Table 4.

Table 3. Geometric Randomization Parameters per Shape Class

Shape Class	Parameter	Units	Range
Sphere	Mesh Segments	[-]	18
Cylinder	Radius	[m]	$U(0.5, 1.2)$
	Height	[m]	radius $\cdot\, U(2.5, 7.5)$
CylinderDome	Dome Offset	[m]	$U(0.7, 2.5)$
Cone	Radius	[m]	$U(0.4, 1.2)$
	Height	[m]	$U(1.5, 5.0)$
CubeSat	Scale factor	[m]	$U(0.8, 1.3)$
FlatPanel	Length	[m]	$U(0.8, 3.3)$
	Width	[m]	$U(0.5, 2.5)$
	Thickness	[m]	$U(0.01, 0.07)$
TruncatedCone	Base Radius	[m]	$U(0.5, 1.3)$
	Top/Bot Ratio	[-]	$U(0.2, 0.9)$
	Height	[m]	$U(2.0, 7.0)$

Table 4. Geometric Randomization Parameters for Asteroid Classes

Shape Class	Parameter	Units	Range
Asteroid Prolate	Elongation	[-]	$U(3.0, 5.5)$
	Flatness	[-]	$U(0.45, 0.85)$
	Roughness	[-]	$U(0, 0.08)$
Asteroid Oblate	Elongation	[-]	$U(0.9, 1.35)$
	Flatness	[-]	$U(0.15, 0.40)$
	Roughness	[-]	$U(0, 0.12)$
Asteroid Rough	Elongation	[-]	$U(1.5, 3.0)$
	Flatness	[-]	$U(0.50, 0.90)$
	Roughness	[-]	$U(0.20, 0.40)$
Asteroid Complex	Elongation	[-]	$U(1.5, 3.5)$
	Flatness	[-]	$U(0.40, 0.90)$
	Roughness	[-]	$U(0.35, 0.60)$

3.1.2 Network Training

With the randomized parameters and the dataset described in section 3.1.1, the network can be trained to learn shape classes.

This process can be illustrated with Table 5 and Figure 8.

Table 5. Model Training Parameters and Performance Metrics

Category	Parameter	Value
Data Partition	Total Dataset Size	15000 samples
	Training Set	80% (12000 samples)
	Validation Set	20% (3000 samples)
Training Schedule	Total Epochs	60
	Total Iterations	2760
	Validation Frequency	Every 50 iterations
Learning Rate (LR)	Initial LR	0.001
	LR Drop Factor	0.2
	LR Drop Period	Epoch 36 (60% of epochs)
Results	Validation Accuracy	54.10%
	Learning Time	343 min 34 sec

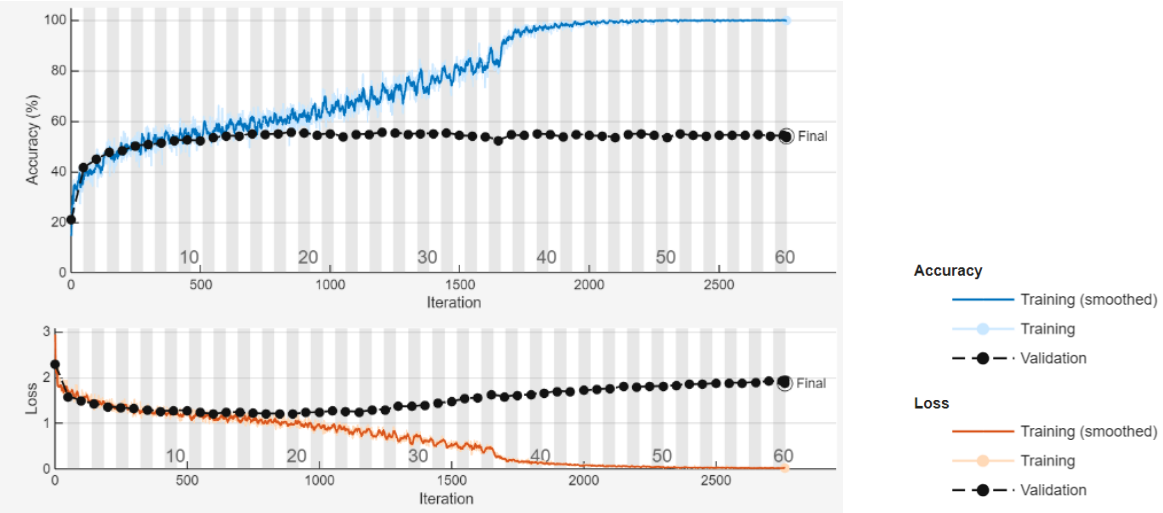


Figure 8. ResNet-1D Training for Accuracy and Loss Function

3.1.3 ResNet-1D Architecture

Before passing through all the different layers of the network, a z-score normalization is applied to the simulated light curves, normalizing to zero mean and unit standard deviation:

$$m_{\text{znorm}}(t) = \frac{m_{\text{app}}(t) - \mu_{\text{LC}}}{\sigma_{\text{LC}}} \tag{72}$$

$$\mu_{LC} = \frac{1}{T}\sum_{t=1}^{T} m_{\text{app}}(t) \tag{73}$$

$$\sigma_{\text{LC}} = \text{std}(m_{\text{app}}(t)) \tag{74}$$

This normalization removes the absolute magnitude level that encodes information about albedo, size and distance. Thus, the network only received the light curve encoding the shape of RSOs.

The light curves then go through five different stages described in Figure 9. All different filters and layers are defined in Section 2.3.

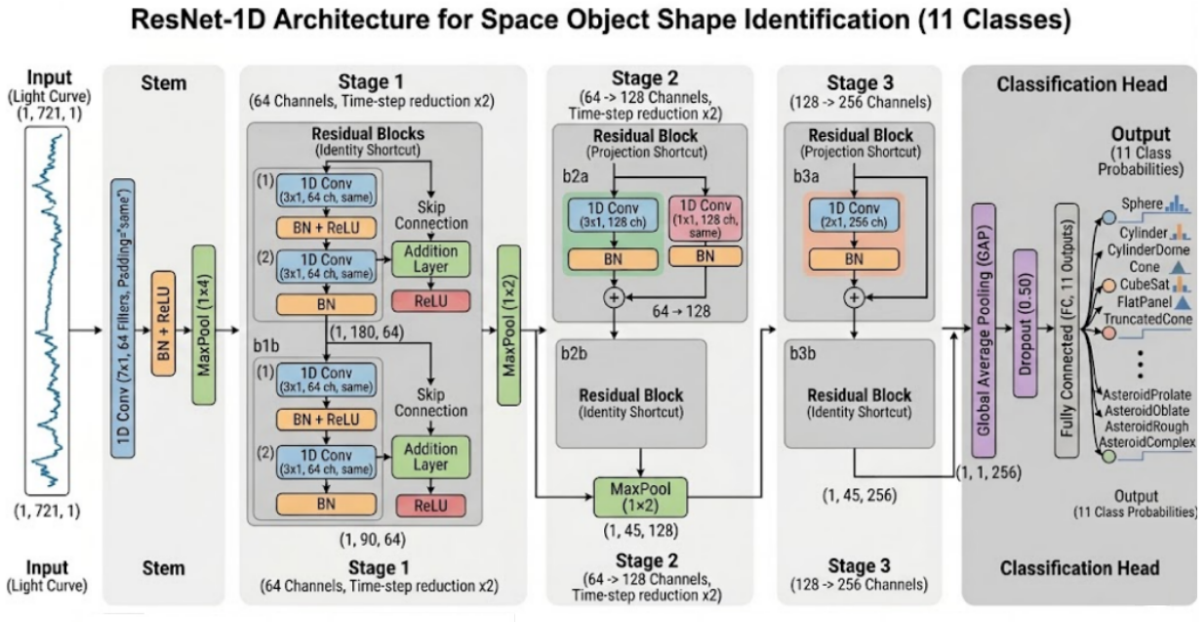


Figure 9. ResNet-1D Architecture

3.2 Attitude Estimation using the GPUKF Algorithm

3.2.1 Prediction and Observation Models

The UKF is commonly used to extract attitude from light curves. As shown by equations (36) and (37), this method requires both a prediction and an observation model. In this paper, the following discrete non-linear system is used:

$$\mathbf{x}_{\text{pred},k+1} = f(\mathbf{x}_{\text{pred},k}) \tag{75}$$

$$m_{\text{app},k} = GP(\mathbf{x}_{\text{obs},k}, \mathscr{D}_{\text{obs}}) \tag{76}$$

with Equations (75) and (76) being respectively the prediction and observation models. $\mathbf{x}_{\text{pred}}$ is the input prediction vector, $\mathbf{x}_{\text{obs}}$ is the input observation vector, m_{app} is the apparent magnitude defined by Equation (10), and $\mathscr{D}_{\text{obs}}$ is the training data set defined Section 3.2.2.

The prediction model f uses both quaternion kinematics (14) and Euler equation of motion (18) to predict the evolution of the attitude. The inertia of RSOs is computed as shown in Section 2.2.2. A geometry simplification, with all fixed positions, is assumed and the initial angular velocity is supposed to be known.

Contrary to the prediction model, the observation model is not clearly defined, since there are uncertainties about the surface parameters when calculating the apparent magnitude of random RSOs. To tackle these uncertainties, the GPR method is used, allowing to describe the non-parametrically model with precision and robustness due to the numerous training data.

3.2.2 Data Training

Let $\mathscr{D}_{obs}$ be the data set used to train the observation model with the GPR:

$$\mathscr{D}_{\text{obs}} = \langle \mathbf{X}_{\text{obs}}, \mathbf{Y}_{\text{obs}} \rangle \tag{77}$$

$$\mathbf{X}_{\text{obs}} = [\mathbf{x}_{\text{obs},1}, \mathbf{x}_{\text{obs},2}, \ldots, \mathbf{x}_{\text{obs},n}] \tag{78}$$

$$\mathbf{Y}_{\text{obs}} = [m_{\text{app},1}, m_{\text{app},2}, \ldots, m_{\text{app},n}] \tag{79}$$

with $\mathbf{X}_{obs}$ the input entry vectors and $\mathbf{Y}_{obs}$ the corresponding output data.

Every input vector is defined as:

$$\mathbf{x}_{\text{obs}} = [\mathbf{q}^T, m_{\text{app,past}}] \tag{80}$$

where $\mathbf{q}$ is the quaternion of the RSO and $m_{\text{app,past}}$ is the mean of the past apparent magnitudes after t_{past} seconds of observation. $m_{\text{app,past}}$ should be included in the input data set to tackle the possibility of a non-unique problem. Indeed, since many different surface parameters and attitude are considered when training the data, it is possible to obtain the same light curves with different combinations. The evolution of the light curve gives information about the change in attitude for specific surface parameters. Thus, it is necessary to consider the mean of the past values of the light curve to make this problem unique [6].

Every possible attitude must also be included in training. To do so, the training data set considers a variation of the principal axis $\mathbf{e}$ and the principal angle Φ throughout the whole space state as represented in Figure 10.

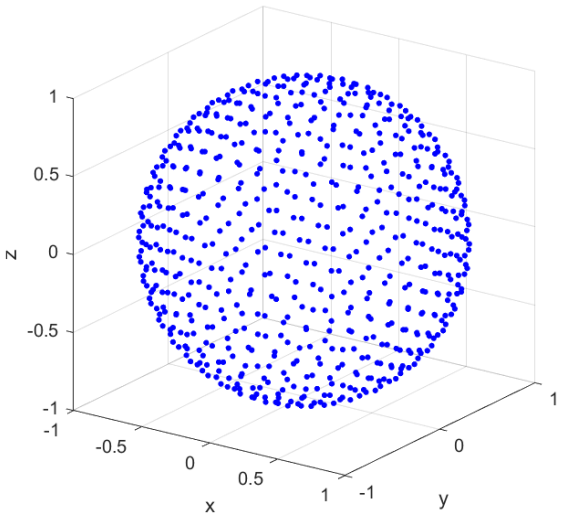


Figure 10. Attitude Space Training Coverage

3.2.3 GPUKF Algorithm

The GPUKF algorithm ([6], [18]) can be summarized in four steps. First, it calculates the sigma points (38)–(40). Since the local attitude error relies on the GRP (15), the sigma points are calculated using these parameters before running through the GP observation model (76) to predict the observed values.

Then, applying the Kalman gain (50) to the process, the posterior estimated state (53) and posterior covariance matrix (54) are updated. Finally, $m_{\text{app,past}}$ is updated with the following formula:

$$m_{\text{app,past},k+1} = \frac{t_{\text{span,past}} m_{\text{app,past},k} + m_{\text{app},k}}{t_{\text{span,past}} + 1} \tag{81}$$

This whole algorithm process is described in Figure 11.

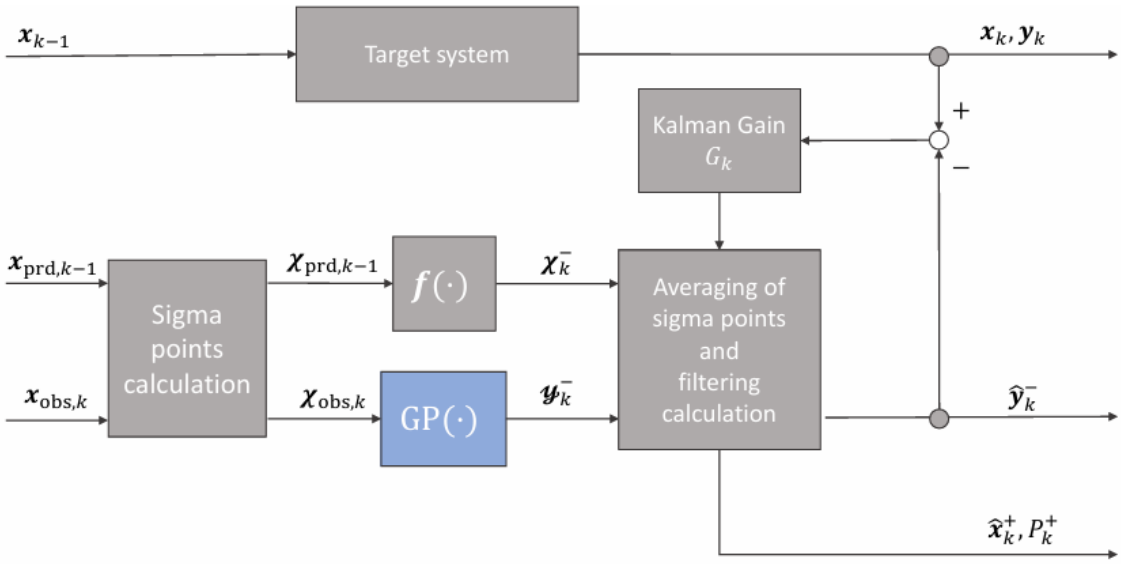


Figure 11. GPUKF Algorithm [6]

Throughout the whole simulation process, the relative position, in Table 6, of the RSO, observer and Sun is assumed to be fixed to represent a geosynchronous orbit scenario.

Table 6. Object-Observer-Sun relative position in the body-fixed inertial frame

Parameter	Value
Object position	$(0,0,0)$ km
Observer position	$(36000,0,0)$ km
Sun position	$(1.2975,-0.3461,-0.6592)\times 10^8$ km

In order to compute the mass and inertia of RSOs, a uniform density of 2700 kg/m^3 is assumed. The other parameters are defined in Sections 3.1.1 and 3.2.2.

4.1 Shape Modeling

Random light curves are computed and associated to a specific class with a probability ranking. It is calculated using the softmax function, defined by Equation (60), giving the top three predictions with the highest probability. The predicted class is the one with the highest probability.

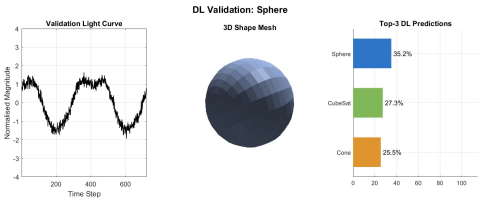


Figure 12. Sphere Class Prediction

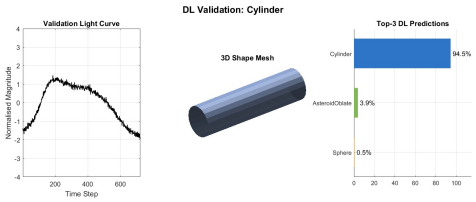


Figure 13. Cylinder Class Prediction

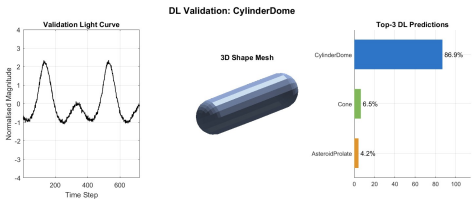


Figure 14. Cylinder Dome Class Prediction

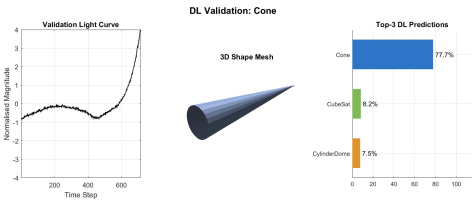


Figure 15. Cone Class Prediction

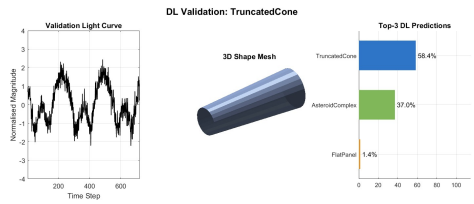


Figure 16. Truncated Cone Class Prediction

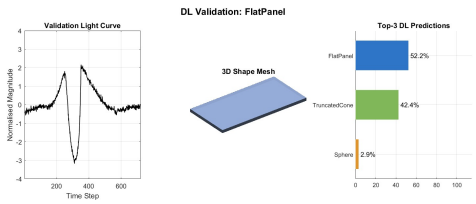


Figure 17. Flat Panel Class Prediction

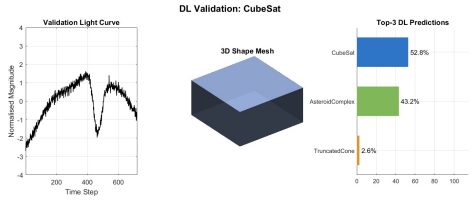


Figure 18. CubeSat Class Prediction

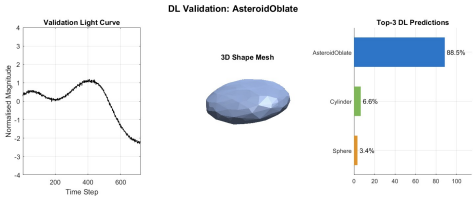


Figure 19. Asteroid Oblate Class Prediction

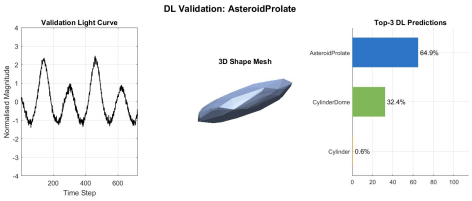


Figure 20. Asteroid Prolate Class Prediction

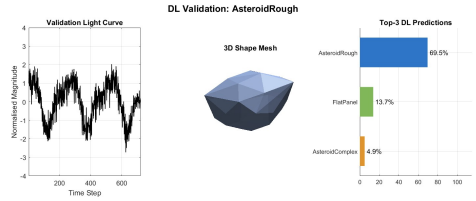


Figure 21. Asteroid Rough Class Prediction

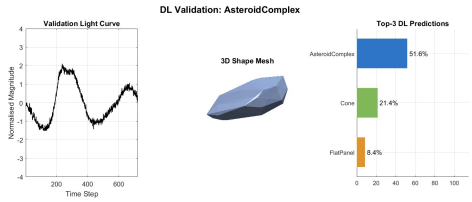


Figure 22. Asteroid Complex Class Prediction

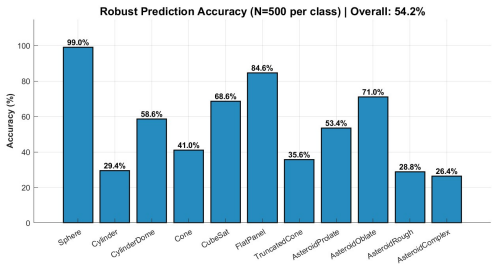


Figure 23. Class Prediction Accuracy

The results show some disparity in the predictions of each class. Indeed, it can be extracted from the network predictions that some classes (Figures 12, 16, 17, and 18) have a dominating probability from the true class, while other classes (Figures 13, 14, 15, 19, 20, 21, and 22) have close probability predictions between the true class and an other class.

Since the light curves and shapes are randomly generated, it can be difficult to establish a specific relation between close classes predictions. However, using Figures 23 and 24, assumptions can be made. First, it can be noticed that, while having close probability guesses between two classes, the sphere and flat panel models have really high correct prediction overall. On the opposite, cylinder, rough and complex asteroid models have a much lower prediction accuracy.

Figure 24, gives more details about the classes prediction with a total of 1100 samples. The accuracy tendency is matching the one in Figure 23 but adds as well, in case of a false prediction, which class was wrongly modeled.

The confusion matrix shows a good performance of the ResNet-1D in identifying geometries with unique photometric signatures. For example, the sphere, with its rotational symmetry, will produce a low variance light curve regardless of its attitude. The flat panel light curve will also be characterized by its sharp glints, easily recognizable for the network. However, confusion can be observed between classes with low accuracy percentage. It is especially noticeable between cylinder, cylinder dome, and cone classes, and between rough and complex asteroid classes. This can be explained by a mathematically similar geometry and reflective area between the different classes when facing a specific angle.

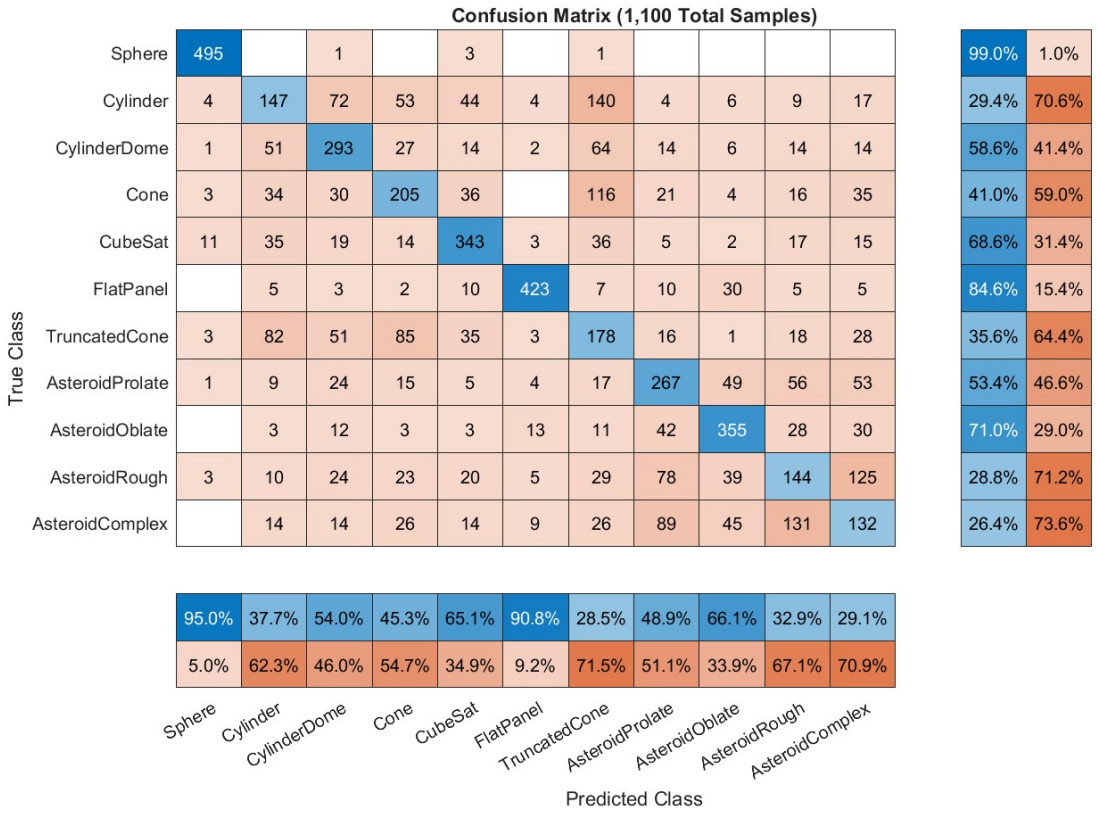


Figure 24. Confusion Matrix

4.2 Attitude Estimation

The initial Euler angles error is set as $(\phi_{\text{err}}, \theta_{\text{err}}, \psi_{\text{err}}) = (-10°, 5°, -10°)$. Table 7 provides the average Root Mean Square Error (RMSE) results for the attitude estimation.

For most of the shapes, the GPUKF results show a convergence trend towards a steady-state more or less close to a null attitude error. While for some shapes like the sphere, the cubesat, or the complex asteroid (Figures 25, 31, and 35), the error stays close to zero during the whole simulation time, some other shapes like the oblate and prolate asteroids (Figures 32 and 33) have higher average RMSE but end up with an almost null error for Roll angles.

It can be noticed that the sphere (Figure 25) has the lowest overall error due to its rotational symmetry that provides a low-variance light curve easily recognizable whatever the attitude is. Other shapes like the cylinder, the cylinder dome or the cone (Figures 26, 27, and 28) exhibit a much higher Roll RMSE than the Yaw and Pitch RMSE. These shapes all have a longitudinal symmetry axis which, while rotated around, will result in minimal changes of the observed reflected areas, making it difficult for the filter to rely on the photometric data alone. An extreme Yaw RMSE for the prolate asteroid (Figure 33) can be identified, suggesting the struggles of the filter for complex elongated shapes. These divergence tendencies can also be associated with matching error of the predicted and observed light curves, particularly visible for the cone and the prolate asteroid.

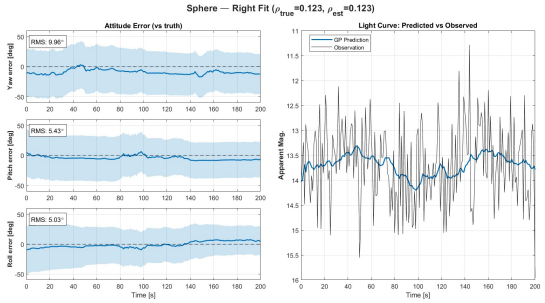


Figure 25. Sphere GPUKF Results

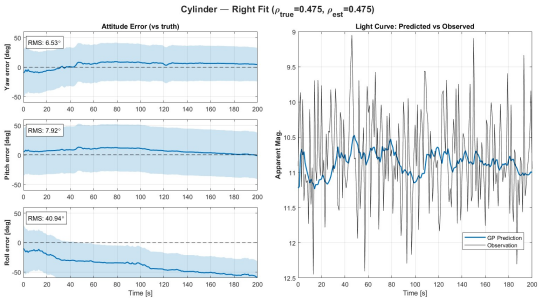


Figure 26. Cylinder GPUKF Results

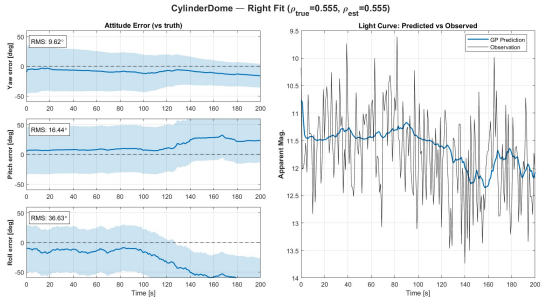


Figure 27. Cylinder Dome GPUKF Results

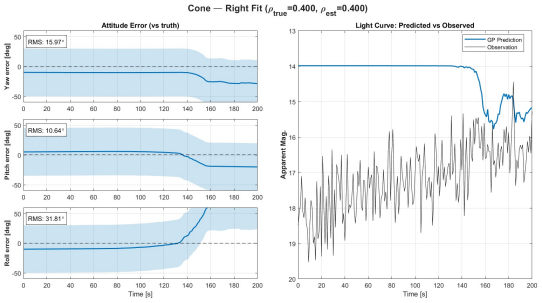


Figure 28. Cone GPUKF Results

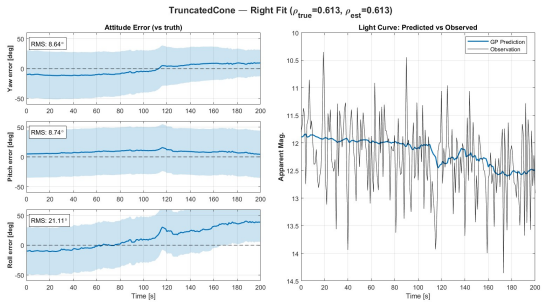


Figure 29. Truncated Cone GPUKF Results

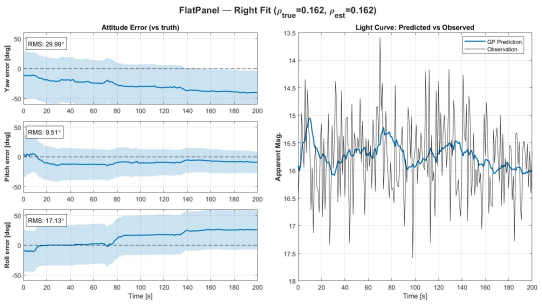


Figure 30. Flat Panel GPUKF Results

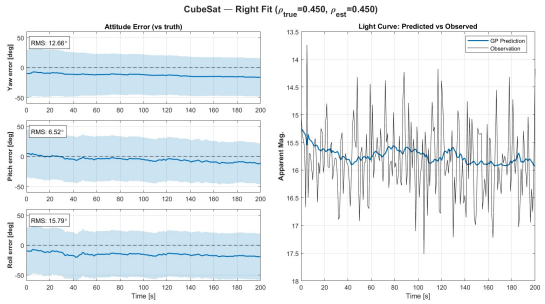


Figure 31. CubeSat GPUKF Results

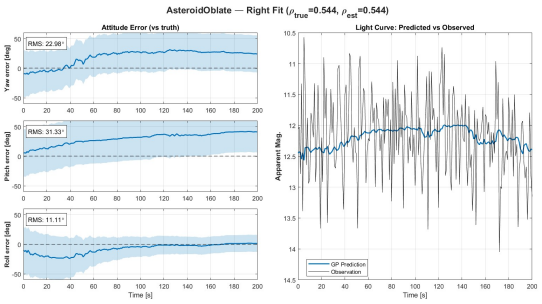


Figure 32. Asteroid Oblate GPUKF Results

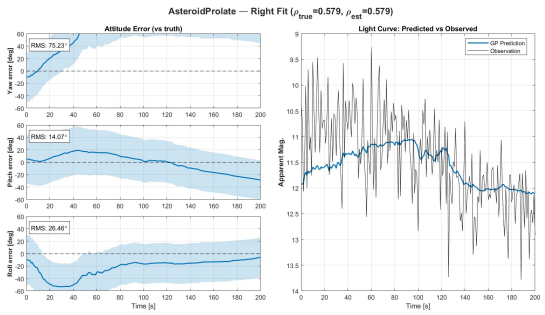


Figure 33. Asteroid Prolate GPUKF Results

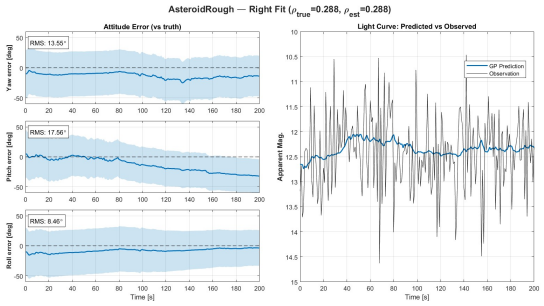


Figure 34. Asteroid Rough GPUKF Results

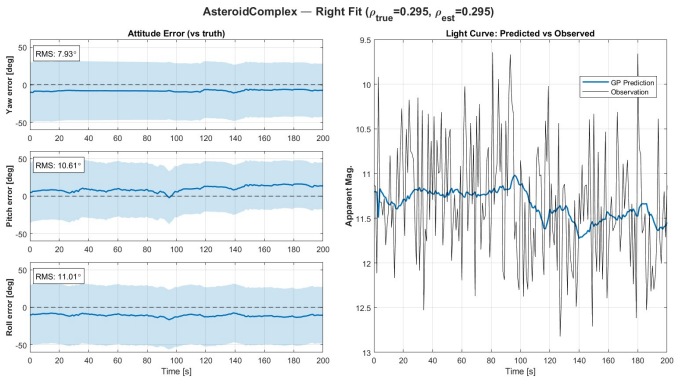


Figure 35. Asteroid Complex GPUKF Results

Table 7. Average RMSE Attitude Estimation Results for a Right-fit ρ_d

Shape Class	**Yaw RMSE (°)**	**Pitch RMSE (°)**	**Roll RMSE (°)**
Sphere	9.96	5.43	5.03
Cylinder	6.53	7.92	40.94
CylinderDome	9.62	16.44	36.63
Cone	15.97	10.64	31.81
CubeSat	12.66	6.52	15.79
FlatPanel	29.99	9.51	17.13
TruncatedCone	8.64	8.74	21.11
AsteroidProlate	75.23	14.07	26.46
AsteroidOblate	22.98	31.33	11.11
AsteroidRough	13.55	17.56	8.46
AsteroidComplex	7.93	10.61	11.01

5 DISCUSSION

To investigate the GPUKF's robustness against uncertainties in surface properties, three scenarios were tested. Section 4.2 introduces the matching true and expected diffusive reflectance case. Appendix A and appendix B deal respectively with the case of an under-fit and an over-fit estimation. Figure 36, summarizing the results of Tables 7, 8, and 9, compares the average RMSE for the three scenarios.

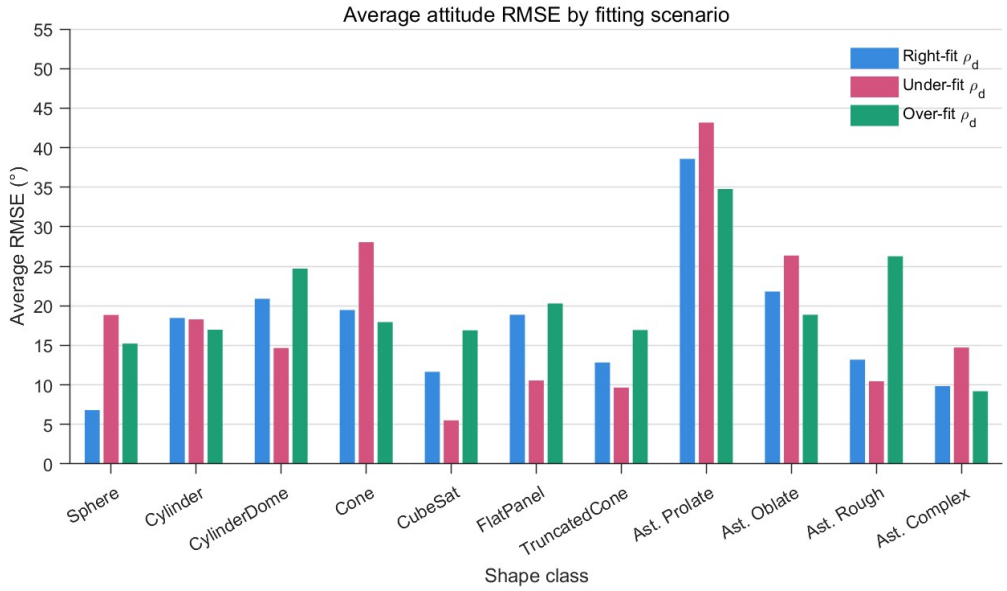


Figure 36. Average RMSE Comparison

This graphic reveals the sensitivity of the GPUKF's surface uncertainty towards shape classes.

For shapes with rotational or longitudinal symmetry, such as the sphere, the cylinder or the cone, the right-fit scenario seems to lead to lower RMSE. While it is particularly visible for the sphere, it remains close for the other shapes. Indeed, the sphere is particularly sensitive to diffusive reflectance mismatch with its Yaw RMSE for the under-fit case, and its Pitch RMSE for the over-fit case, going respectively from 9.96° (Figure 25) to 28.61° (Figure 37), and from 5.43° (Figure 25) to 19.69° (Figure 48). The cylinder like shapes (Figures 26, 27, 28, 29, 38, 39, 40, 41, 49, 50, 51, and 52) have a Roll RMSE that remains significantly high, due to their geometry, not being impacted by a change of diffusive reflectance.

For shapes with specific and richer photometric signature (Cylinder dome, Cubesat and Flat panel), the under-fit scenario performs the best (Figures 39, 43, and 42). An underestimated value of ρ_d helps softening the predicted light curve, which reduces prediction errors that would cause the UKF to overreact. On the opposite, the over-fit scenario introduces brightness inconsistencies to the model that leads to higher RMSE overall.

Asteroid classes provide various behavior. It is clear that the prolate asteroid is the most challenging shape, with extreme Yaw RMSE of 75.23° and 91.65° for both right-fit (Figure 33) and under-fit (Figure 45) scenarios. The over-fit (Figure 56) scenario leads to a diminution of this error with a value of 38.68°, but also have a higher Roll RMSE (48.58°). Except from the over-fit for rough asteroids (Figure 57), the other asteroid shapes seems relatively robust to the diffusive reflectance mismatch. This can be explained by the surface irregularities producing diverse brightness patterns that gives enough information for the GP model.

6 CONCLUSION AND FUTURE WORK

This paper details a viable framework to characterize RSOs with unknown shapes from photometric data. A ResNet-1D structure built for shapes identification achieved such a classification with an overall accuracy of 54.2%, pointing out a disparity of accuracy when predicting different classes. The network performed well for shapes presenting unique features, while having low accuracy and higher confusion for similar structured or more complex shapes. The integration of GPR, as an observation model, within an UKF allowed for attitude estimation with remaining surface properties uncertainty.

This paper still relies on specific assumptions, which could be further explored to improve the model. The density, supposed to be known, could later be included within the training model. The current network is also training for simple shapes and a limited amount of data. Thus, a more accurate shape model for more complex classes could be studied.

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APPENDIX A: UNDER-FIT ρ_D CASE

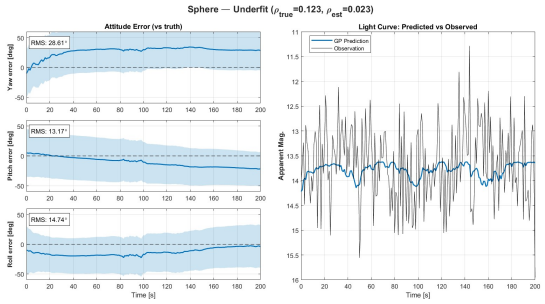


Figure 37. Sphere GPUKF Results

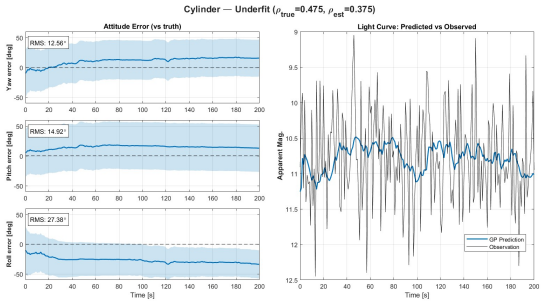


Figure 38. Cylinder GPUKF Results

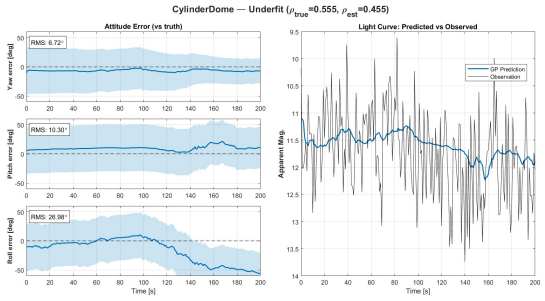


Figure 39. Cylinder Dome GPUKF Results

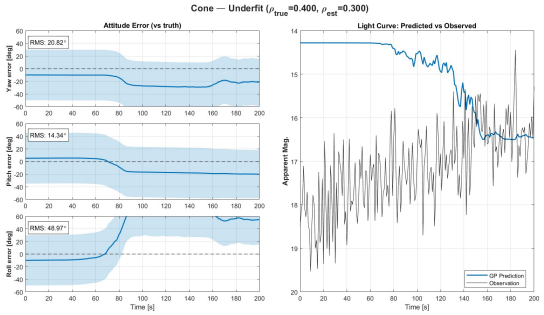


Figure 40. Cone GPUKF Results

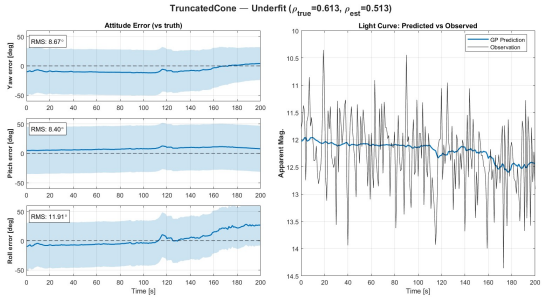


Figure 41. Truncated Cone GPUKF Results

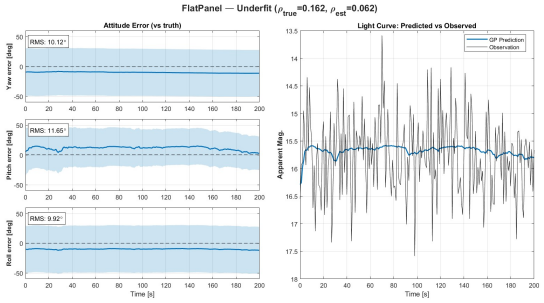


Figure 42. Flat Panel GPUKF Results

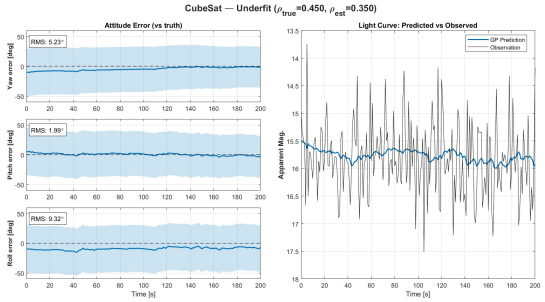


Figure 43. CubeSat GPUKF Results

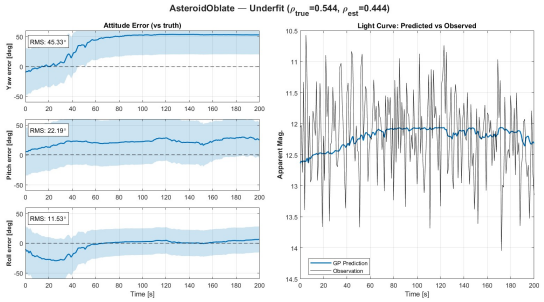


Figure 44. Asteroid Oblate GPUKF Results

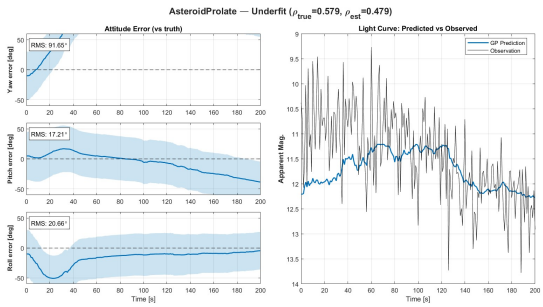


Figure 45. Asteroid Prolate GPUKF Results

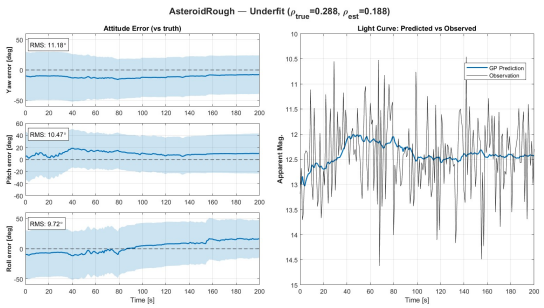


Figure 46. Asteroid Rough GPUKF Results

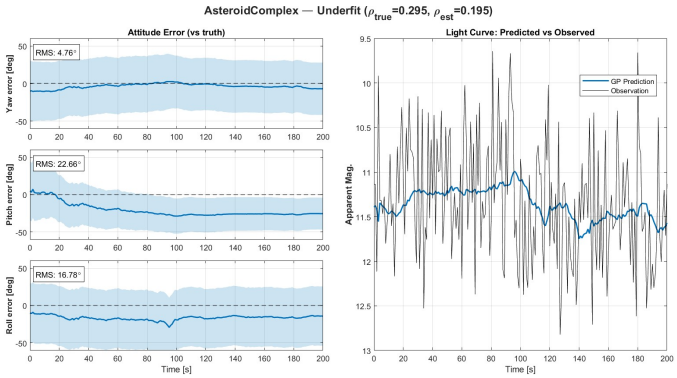


Figure 47. Asteroid Complex GPUKF Results

Table 8. Average RMSE Attitude Estimation Results for an Under-fit ρ_d

Shape Class	**Yaw RMSE (°)**	**Pitch RMSE (°)**	**Roll RMSE (°)**
Sphere	28.61	13.17	14.74
Cylinder	12.56	14.92	27.38
CylinderDome	6.72	10.30	26.98
Cone	20.82	14.34	48.97
CubeSat	5.23	1.99	9.32
FlatPanel	10.12	11.65	9.92
TruncatedCone	8.67	8.40	11.91
AsteroidProlate	91.65	17.21	20.66
AsteroidOblate	45.33	22.19	11.53
AsteroidRough	11.18	10.47	9.72
AsteroidComplex	4.76	22.66	16.78

APPENDIX B: OVER-FIT ρ_D CASE

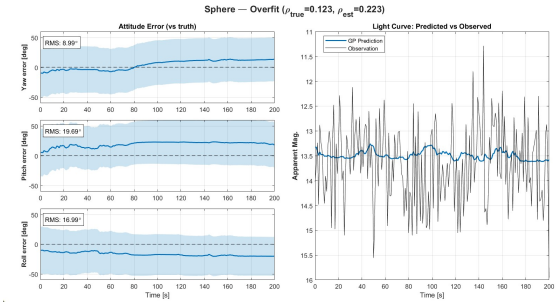


Figure 48. Sphere GPUKF Results

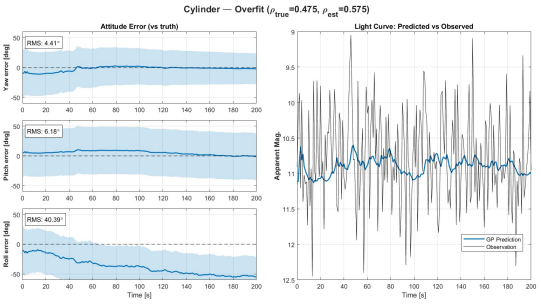


Figure 49. Cylinder GPUKF Results

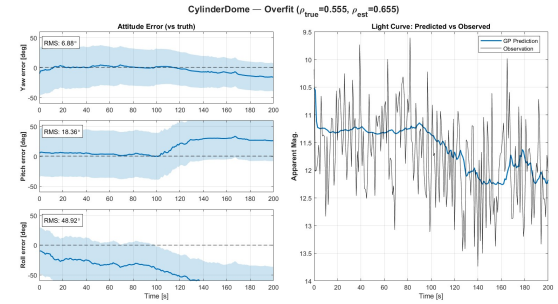


Figure 50. Cylinder Dome GPUKF Results

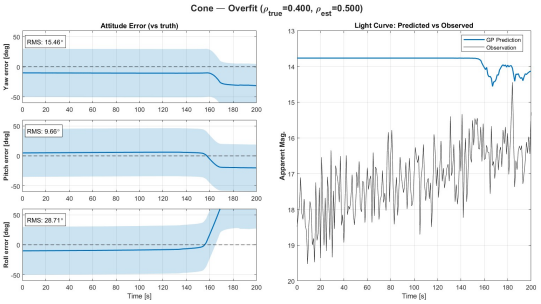


Figure 51. Cone GPUKF Results

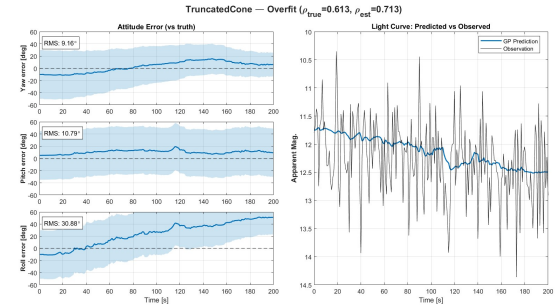


Figure 52. Truncated Cone GPUKF Results

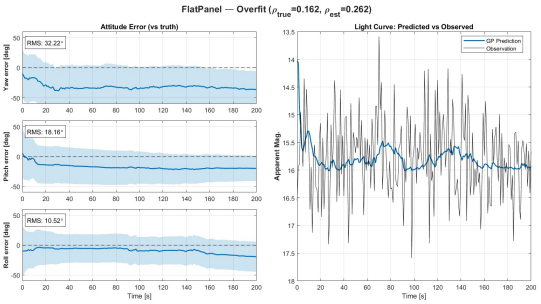


Figure 53. Flat Panel GPUKF Results

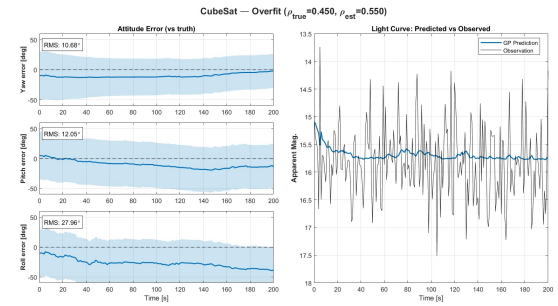


Figure 54. CubeSat GPUKF Results

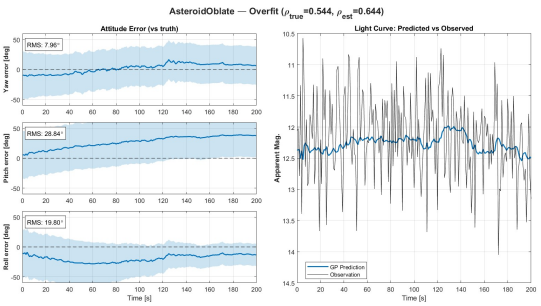


Figure 55. Asteroid Oblate GPUKF Results

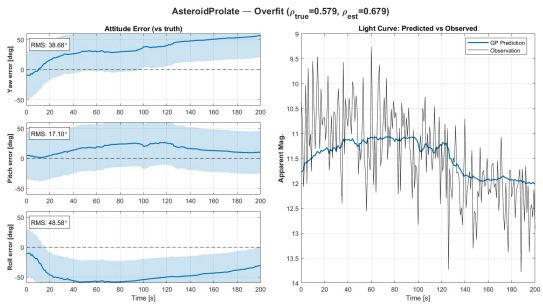


Figure 56. Asteroid Prolate GPUKF Results

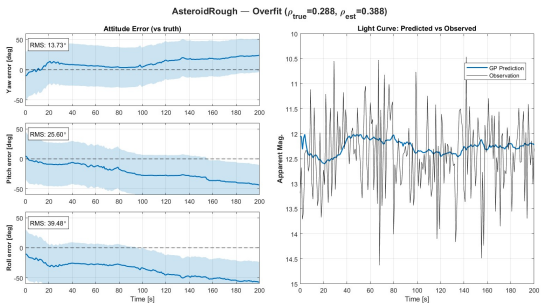


Figure 57. Asteroid Rough GPUKF Results

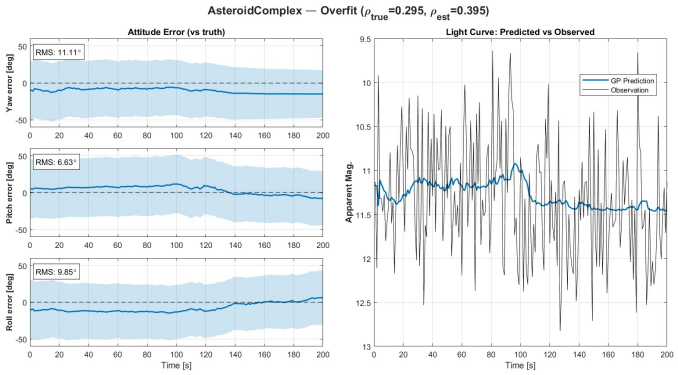


Figure 58. Asteroid Complex GPUKF Results

Table 9. Average RMSE Attitude Estimation Results for an Over-fit ρ_d

Shape Class	**Yaw RMSE (°)**	**Pitch RMSE (°)**	**Roll RMSE (°)**
Sphere	8.99	19.69	16.99
Cylinder	4.41	6.18	40.39
CylinderDome	6.88	18.36	48.92
Cone	15.46	9.66	28.71
CubeSat	10.68	12.05	27.96
FlatPanel	32.22	18.16	10.52
TruncatedCone	9.16	10.79	30.88
AsteroidProlate	38.68	17.10	48.58
AsteroidOblate	7.96	28.84	19.80
AsteroidRough	13.73	25.60	39.48
AsteroidComplex	11.11	6.63	9.85